Tactix public Tseitin cycle-breaking action
Certified reasoning checkpoint

Development report for Andrew

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Outcome

Tseitin cycle breaking now reaches the public controller boundary. Commit `46a7db1cd` registers X16 at append-only rule ID 38, disabled reference StepId 27, and C parameter tag 41 in API version 7. Its parent `2921c784d` repairs the common native structured-producer lifecycle. Explicit programs and an opt-in random producer use the same typed executor; DefaultPolicy remains disabled pending fuzz, recovery, and regression evidence.

This is a correctness and architecture checkpoint. No target, WV/WV2, Tier-D, fuzz, or regression campaign was launched. The saved `d170b4a5` comparison was not rerun.

Controller-visible component choice

The bounded native census returns every eligible connected XOR component as a separate exact `tcx2:` payload. It does not privately choose which component the controller receives. The payload commits the stored clauses, native proof identities, variable mapping, root image, admitted XOR groups, incidence, available-edge cycle, X7 witness, transaction, and limits. A distinct `tcd1:` identity records semantic duplicate state in one Kernel-bound session.

Apply recaptures the formula, selects only the named component, rebuilds the complete plan, and exact-matches it before mutation. Success publishes one cycle-breaking unit. Propagation remains a separate action. A fresh census at the next propagated root can expose another component or an alternative cycle. Root-fixed off-cycle edges are authenticated; selected cycle edges must remain unassigned.

The present domain is conservative: any unclaimed stored clause touching a component edge excludes the component, even when root-satisfied. This physical residual boundary is stronger than the abstract X7 condition and remains an explicit N1 review item.

Boundary	Contract	State
Offer	Mutation-free complete census; one exact payload per component	pass
Apply	Fresh component/root/source rebuild and exact match	pass
Proof	No-proof or native DSR; unsupported formats decline first	pass
Replay	C++ serialization and additive C v7 deep-copy	pass
Default scheduling	Explicit disabled row; production admission unmeasured	pending
Recovery and gates	Fixed 44, WV/WV2, B28+C5, and Rank	pending

Proof, provenance, and retry behavior

Native DSR emits two temporary helpers, one RUP unit, two deletions, and one proof-only variable. It adds no branchable auxiliary and needs no model reconstruction witness. DRAT, LRAT, and PBP decline before mutation. This keeps the first public route in the default DSR proof vocabulary without adding a rule-specific pseudo-Boolean path.

The prebuilt JSONL record binds action and semantic identities, typed limits, the mapping and authenticated root units, exact source occurrences and proof IDs, the cycle and witness, proof events, installed unit, projection/deletion state, and exact effort and proof volume. Failed publication neither consumes the duplicate key nor closes the certificate boundary.

Applied, duplicate, and complete no-structure outcomes consume the offer. Invalid, inapplicable, unsupported-proof, and stale results discard it until context changes. Exhaustion and interruption retain it while suspending eligibility; an explicit profile-only change retries it without rediscovery. Formula, session, or owner changes discard it. There is no fallback or rollback.

Validation

Independent review found no core soundness, reachability, ABI, lifecycle, or proof blocker. A separate fresh Debug build passed 10/10 focused tests covering the public action, admission, discovery, live adapter, planning, transaction, journal, API instances, all online actions, and inprocessing programs.

The journal test independently checks ASCII and binary DSR through the proof oracle with every-lemma checking. It also verifies a repeated two-action, propagated-root proof with forward `dsr-trim`. The action test covers exact proof accounting, unsupported modes, stale and corrupt identities, repeated root discovery, alternative-cycle recovery, typed replay, random reach, and C v7 compatibility and deep-copy. Focused ASan/UBSan action and discovery tests passed 2/2 with leak detection disabled because LSAN is unavailable under the environment's tracing.

Catalog, proof-rule synchronization, effort governance, policy boundaries, configuration wiring, section references, progress and document-size checks pass. The feedback inventory contains 736 native fields and no wall timers. A synthetic one-unit CLI smoke confirms the stable Tseitin attempt counter used by the fuzz reach ledger; it is not a benchmark campaign.

N1–N4 and next work

N1 now has public exact component selection, repeated-root discovery, and alternative-cycle recovery, with broad public-sequence coverage still open. N2 has focused checked DSR and no-proof evidence, while broad both-policy fuzz and target checking remain open. N3 is unchanged because the rule is disabled in the reference default. N4 has typed scheduling, raw effort, replay, and retry mechanics, but no accepted interval, allowance, ordering, or default admission.

Next require effective default/random fuzz reach. When benchmark execution is authorized, measure the saved fixed-44 and LymphoSAT-owned WV/WV2 targets, then run B28+C5, Rank1–50, WV100, and WV2–89 with the 1,800-second proof-check cap and the completed-proof slowdown rule. No target or formal gate is relabeled complete by this checkpoint.

The detailed record is the repository's technical checkpoint.